

Assignment #3

Homework Submission Guidelines

1. **Due date: 21.12.2025**
2. The assignment can be done in pairs
3. Answers can be submitted either in English or Hebrew
4. HW submission should be done via moodle in the corresponding area (by **only** one of the students)
5. Late submission penalty (**5% a day**) for submitting after the assignment's due date
6. Questions / clarifications and more in the dedicated discussion sub-forum in Piazza

Dry part (50%)

Language Models (20%):

1. (4%) Given the following documents:
D1 : {a a a d}
D2 : {a a b c}
D3 : {a d d d}
D4 : {b d d d}
And a query $q : \{a d\}$. Rank the 4 documents with respect to the query (in descending order) using KL Divergence. Using Dirichlet for smoothing with the Dirichlet prior set to 1000. Detail your calculations.
2. (4%) Prove that JM (Jelinek Mercer) smoothed language model yields a valid probability mass function. Specify all the requirements and assumptions of this proof.
3. (4%) Show that the ranking induced using the -CE (negative Cross Entropy) score between a query and a document is equivalent to that induced using the query likelihood method. Specify your assumptions.
4. (4%) KL divergence is an asymmetric divergence measure, which measures how different two probability distributions are from each other. In our course we use it to measure how different the probability distribution M_d is at modeling M_q . Suggest a **symmetric** and **non-negative** measure for comparing query and document language models based on KL divergence.
5. (4%) Given a set of terms marked as relevant to an information need and a user's query (there is no overlap between the query terms and the set of relevant terms). Show, using formulas and explanations, how the language model that is induced from the original query can be expanded using these terms. The updated query model should place greater emphasis on the original query terms.

Relevance Models (15%):

1. Given the query q : "cat dog" and the following list of initially retrieved documents:

d_1 : cat dog cow pig horse

d_2 : the cat and the dog are playing together

d_3 : cat cat cat cat cat cat

(5%) Induce a query model using the RM3 relevance model. Use $\beta = 0.3$ and assuming that $p(q|d)$ is constant and that an MLE is used for the document language model.

2. (5%) Mention 2 ways that can be used to prevent query drift in models that are based on pseudo relevance feedback. Limit your answer to 3 lines.

3. (5%) Suggest two **different** methods for measuring the similarity between two **short** queries. Elaborate and explain the methods and the reason that each was chosen.

Passage Retrieval (10%)

The passage retrieval task is ranking passages of documents by their relevance to the information need expressed by a query.

A passage is any sequence of text in a document which is usually much shorter than the entire document length.

The idea is to estimate the relevance according to the probability of generating a passage g given the query q , expressed as $p(g|q)$. Instead of directly estimating this probability, we use Bayes rule and passages can be ranked using the query-likelihood approach: $p(q|g)$. Note that we assume that the prior $p(g)$ is uniform and thus can be removed.

$p(q|g)$ can now be estimated using the standard language-model-based approach. For each passage a language-model is inferred.

Your task is to suggest **3** different approaches (**that were not taught in class**) to estimate $p(q|M_g)$, where M_g is the passage model.

Tips:

1. In your solution you should address the vocabulary mismatch problem between the terms used in the query and in the short relevant passages.
2. Use the document that contains the passage and the collection for smoothing.
3. Each suggestion should result in a **valid** language model.

Elaborate and detail all of your notations, free parameters, equations, etc...

Diversification (5%)

The diversification problem in ad-hoc document retrieval refers to the challenge of providing a diverse set of relevant documents in response to the query. Instead of simply retrieving a list of most highly ranked documents by some retrieval method, the system aims to present a varied selection of documents that cover different aspects of the query and that do **not overlap** with each other as much as possible. This helps to offer more comprehensive results.

Your task is to propose a diversification method that considers the document's relevance score to the query and the level of redundant content compared to the

documents already retrieved. You can select any retrieval method and must provide equations and explanations to support your solution.

Wet part – Kaggle Challenge (50%)

Description

This homework assignment is presented in the format of a Kaggle-style competition. Your goal is to develop a system that generates the best possible short answers to a set of questions, using information retrieved from Wikipedia articles. Your model should return concise, accurate responses based on relevant content from the corpus. **You should return a single answer per question.**

What am I predicting?

For each question, your task is to retrieve a set of relevant passages (i.e., context) from the Wikipedia corpus and provide them as input to the LLM (Llama-3.2-1B-Instruct), which will then extract or generate the correct short answer based on the provided context. You are allowed to modify the provided system and user prompts, but you may not change the LLM (Llama-3.2-1B-Instruct) or the underlying Wikipedia corpus.

Important note: you are not allowed to change the LLM version, the corpus, or the Pyserini index from which you should perform retrieval.

Evaluation

Submissions are evaluated using the average F1 score over all questions. The evaluation code compares each prediction to all acceptable answers (ground truths), normalizes them (lowercasing, punctuation removal, article removal, and whitespace fix), and returns the maximum F1 score over all ground truths for that question.

Expected input format:

- solution: a DataFrame with columns: ['id', 'answers'], where 'answers' is a JSON-style string list (e.g., '["answer1", "answer2"]')
- submission: a DataFrame with columns: ['id', 'prediction'], where 'prediction' is a JSON-style string list that consists of a single prediction.

Your prediction file should be a CSV with two columns: id and prediction. Each row should correspond to a question in test.csv, with 'id' matching the question ID and 'prediction' containing your model's answer.

Grading

The competition will encourage innovation, will provide valuable experience in building end-to-end information retrieval systems, allowing you to see firsthand how various retrieval techniques can influence the relevance and quality of search results. your grade will be determined based on **the quality of your system** and based on **creativity and knowledge of the suggested ranking model. As such, you are required to submit the code of the most effective run and a detailed description of your approach, models, techniques, etc.**

Prerequisites

Before you begin working on this assignment, please make sure you have access to the Llama-3.2-1B-Instruct model on Hugging Face. This model requires authentication, so you must have the following:

1. A Hugging Face Account

If you do not already have one, create a free account at:

<https://huggingface.co/join>

2. Accepted Access to the Model

Some models on Hugging Face require you to request access before use.

Go to the model page for Llama-3.2-1B-Instruct and click "Request Access".

Approval is usually automatic but may take a few minutes.

3. A Hugging Face Access Token

You will need a personal access token to authenticate programmatically.

Steps:

1. Log in to your Hugging Face account.
2. Navigate to Settings → Access Tokens.
3. Click "New token", give it a name (e.g., llama32_token), and select Read permissions.
4. Copy the token somewhere safe — you will use it in your code and/or Kaggle notebook.

4. Configure Authentication in Your Environment

You must authenticate before loading the model.

For example, in Python:

```
from huggingface_hub import login

login(token="YOUR_HF_TOKEN")
```

How to join the challenge:

Follow this link for registration and more information:

<https://www.kaggle.com/t/6b24ce8d45fa4a8294f82619ab4359b7>

*** Make sure your submissions are with a unique group name. Please state your group name in your submitted assignment.

Submission Instructions:

1. A **PDF** file containing all answers to the questions (Dry and the model description for the Wet part).
2. Code files used to run the experiments in the Kaggle Challenge + detailed description of your most effective system in a pdf file.
3. Create a Zip containing the pdf file and the code files.

4. The name of the file as follows:

HW3_Student_1_EMAIL_Student_2_EMAIL.zip